SUPERMARKET SALES ANALYSIS

Business questions:-

-Which city generates the highest revenue?

-Which branch performs best?

-What is the total sales over time (daily/monthly)?

-What time of day has the highest sales?

Which product line sells the most?

-Which product line generates the highest profit (gross income)?

-Which products have low sales but high price?

-Do members vs normal customers spend differently?

-Which gender spends more?

-What is the average quantity per customer type?

-What is the average rating per branch?

-Which product line has the lowest ratings?

-Is there a relationship between rating and sales?

-Which payment method is most used?

-Do people who use certain payment methods spend more?

-Which branch has the highest gross income?

-How does gross margin percentage vary?

-What factors affect profit most (price, quantity, tax

Objective

The goal of this analysis is to understand the factors influencing supermarket sales and profitability across branches.

We also analyze how customer type, product line, and time affect revenue and customer satisfaction.

In [ ]:
```
from google.colab import files
Uploaded = files.upload()
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving supermarket_sales.csv to supermarket_sales.csv
```

In [ ]:
```
data = pd.read_csv("supermarket_sales.csv")
data.head()
```

Out [2]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Total	Date	Time	Payment	cogs	gros margi percentag
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	13:08	Ewallet	522.83	4.761905
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	10:29	Cash	76.40	4.761905
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	13:23	Credit card	324.31	4.761905
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	20:33	Ewallet	465.76	4.761905
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	10:37	Ewallet	604.17	4.761905

In [ ]:
```
data.columns
```

Out [3]:
```
Index(['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender',
       'Product line', 'Unit price', 'Quantity', 'Tax 5%', 'Total', 'Date',
       'Time', 'Payment', 'cogs', 'gross margin percentage', 'gross income',
       'Rating'],
      dtype='object')
```

In [ ]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice ID               1000 non-null   object
 1   Branch                   1000 non-null   object
 2   City                     1000 non-null   object
 3   Customer type            1000 non-null   object
 4   Gender                   1000 non-null   object
 5   Product line             1000 non-null   object
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax 5%                   1000 non-null   float64
 9   Total                    1000 non-null   float64
 10  Date                     1000 non-null   object
 11  Time                     1000 non-null   object
 12  Payment                  1000 non-null   object
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
```

In [ ]:
```
data.describe()
```

Out [5]:

	Unit price	Quantity	Tax 5%	Total	cogs	gross margin percentage	gross income	Rating
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000	1.000000e+03	1000.000000	1000.00000
mean	55.672130	5.510000	15.379369	322.966749	307.58738	4.761905e+00	15.379369	6.97270
std	26.494628	2.923431	11.708825	245.885335	234.17651	6.131498e-14	11.708825	1.71858
min	10.080000	1.000000	0.508500	10.678500	10.17000	4.761905e+00	0.508500	4.00000
25%	32.875000	3.000000	5.924875	124.422375	118.49750	4.761905e+00	5.924875	5.50000
50%	55.230000	5.000000	12.088000	253.848000	241.76000	4.761905e+00	12.088000	7.00000
75%	77.935000	8.000000	22.445250	471.350250	448.90500	4.761905e+00	22.445250	8.50000
max	99.960000	10.000000	49.650000	1042.650000	993.00000	4.761905e+00	49.650000	10.00000

In [ ]:
```
data.shape
```

Out [6]: (1000, 17)

In [ ]:
```
data.isnull().sum()
```

Out [7]:

	0
Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

dtype: int64

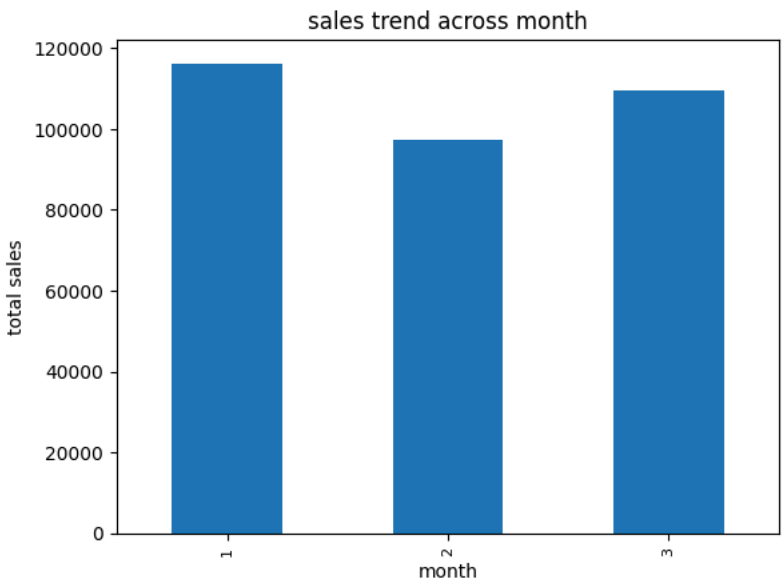
In [ ]:
```
#change the the date to datetime
data['Date'] = pd.to_datetime(data['Date'])
```

```
data['Month'] = pd.to_datetime(data['Date']).dt.month
data['Day'] = data['Date'].dt.day
data["hours"] = data["Date"].dt.hour
```

In [ ]:
```
month_sales = data.groupby(["Month"])["Total"].sum()
print(month_sales)
```

```
Month
1    116291.868
2     97219.374
3    109455.507
Name: Total, dtype: float64
```

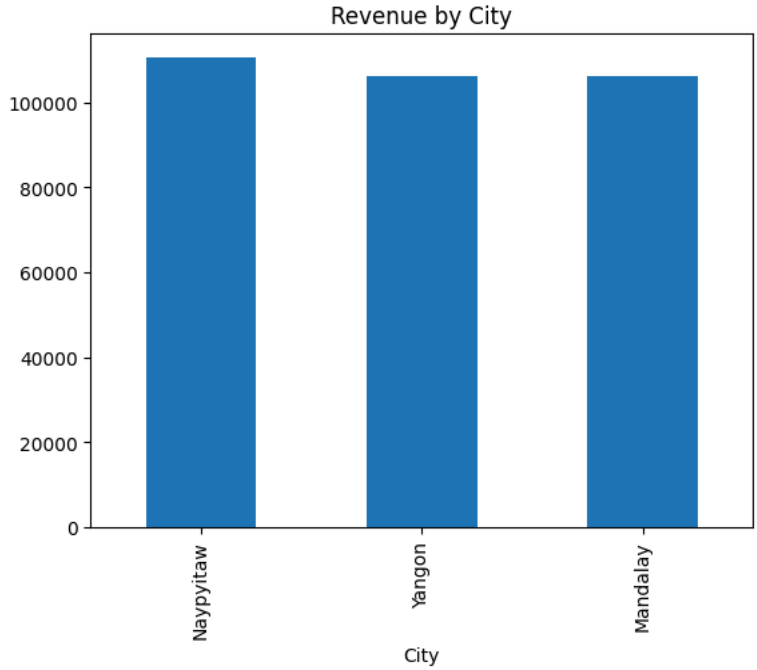
In [ ]:
```
month_sales. plot(kind="bar")
plt.title('sales trend across month')
plt.xlabel('month')
plt.ylabel('total sales')
plt.xticks(size=8)
plt.show()
```



In [ ]:
```
#city with highest sales
city_sales = data.groupby('City')['Total'].sum().sort_values(ascending=False)
print(city_sales)

city_sales.plot(kind='bar')
plt.title('Revenue by City')
plt.show()
```

```
City
Naypyitaw    110568.7065
Yangon       106200.3705
Mandalay     106197.6720
Name: Total, dtype: float64
```



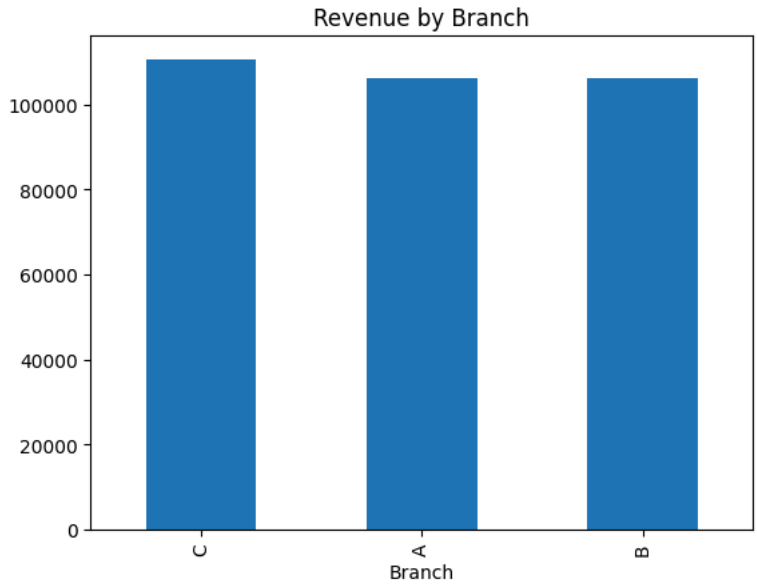
Insight: Revenue by City

- Naypyitaw generates the highest revenue, followed by Yangon and Mandalay.
- This suggests that Naypyitaw has stronger customer demand or higher transaction values.

Recommendation:

The business should focus marketing and expansion strategies in Naypyitaw.

In [ ]:
```
#which branch has highest sales
branch_sales = data.groupby('Branch')['Total'].sum().sort_values(ascending=False)
branch_sales.plot(kind='bar')
plt.title('Revenue by Branch')
plt.xticks(rotation='vertical')
plt.show()
```



Insight

Branch Performance

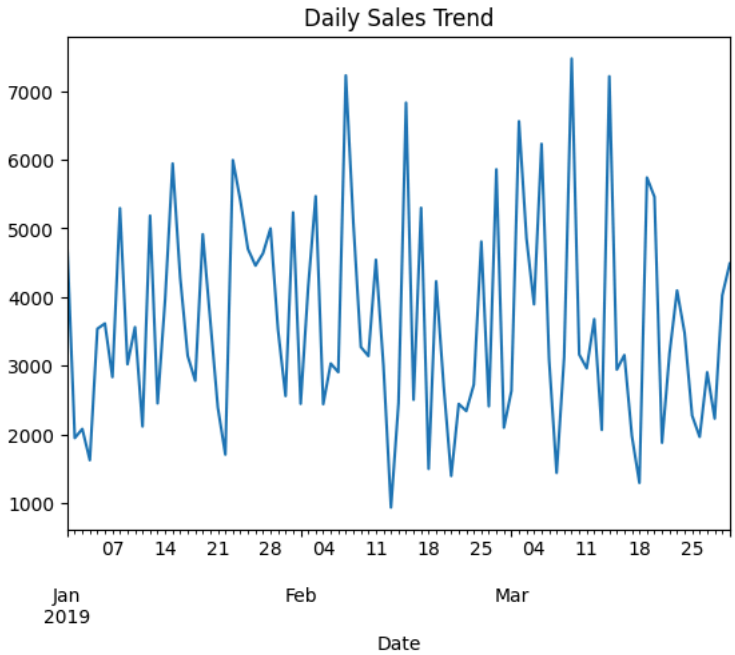
Branch C has the highest revenue among all branches.

This indicates better performance in terms of sales volume or pricing.

Recommendation: Analyze what makes Branch C successful and apply the same strategy to other branches.

In [ ]:
```
#trend of sales over time
daily_sales = data.groupby('Date')['Total'].sum()
```

```
daily_sales.plot()
plt.title('Daily Sales Trend')
plt.show()
```



Insight

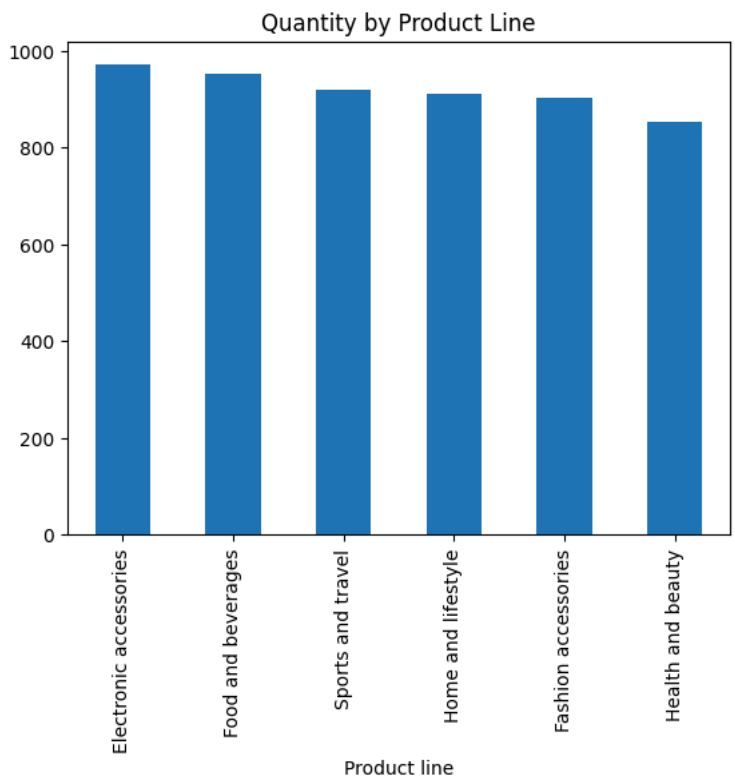
Daily Sales Trend

Sales fluctuate across different days, showing no consistent pattern.

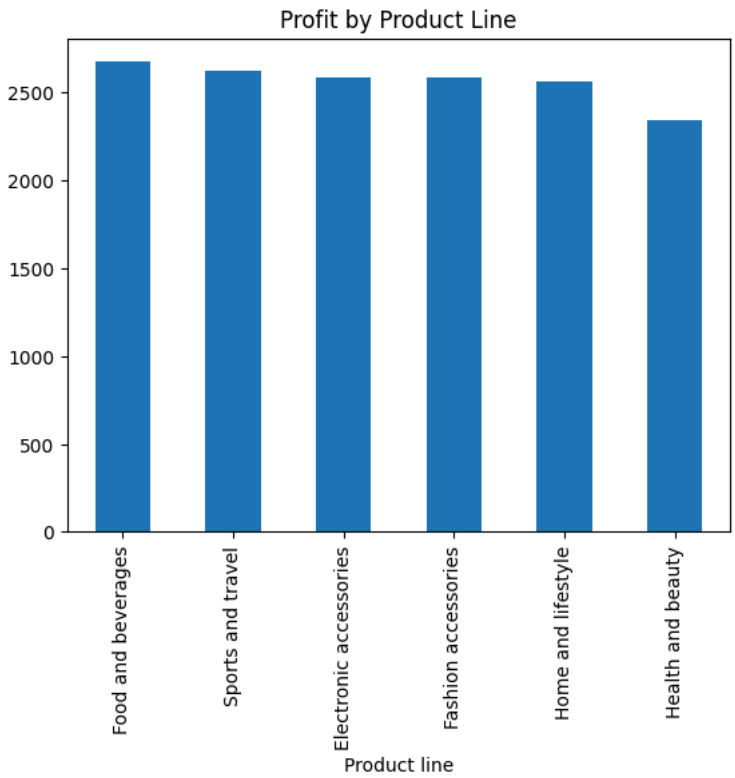
This indicates variability in customer demand.

Recommendation: Introduce promotions on low-sales days to stabilize revenue.

In [ ]:
```
#best selling product
product_sales = data.groupby('Product line')['Quantity'].sum().sort_values(ascending=False)
product_sales.plot(kind='bar')
plt.title('Quantity by Product Line')
plt.show()
```



In [ ]:
```
#Product with high revenue
profit_product = data.groupby('Product line')['gross income'].sum().sort_values(ascending=False)
profit_product.plot(kind='bar')
plt.title('Profit by Product Line')
plt.show()
```



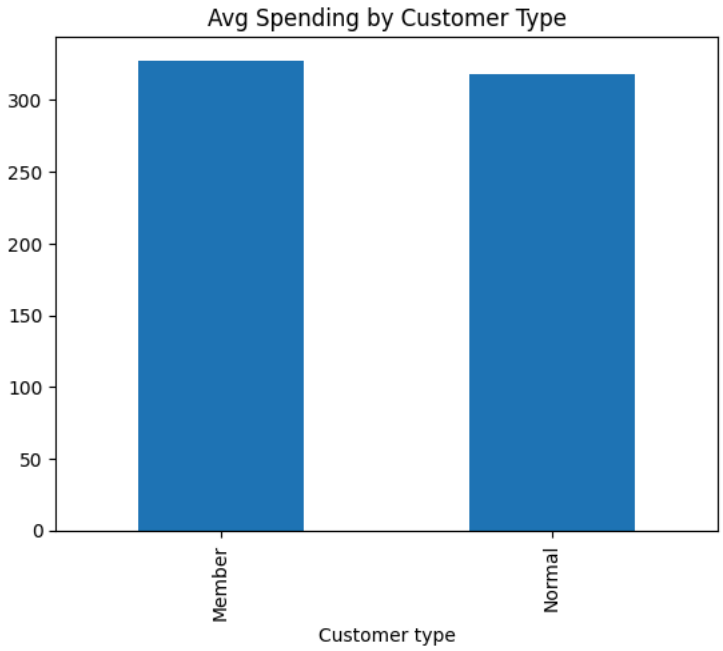
Insight: Product Performance

Electronic accessories are the most sold product line, while food and beverages generate the highest profit.

This shows that high sales volume does not always mean highest profit.

Recommendation: Focus on both high-volume and high-profit products for balanced growth.

In [ ]:
```
#which customer buy more
customer_spending = data.groupby('Customer type')['Total'].mean()
customer_spending.plot(kind='bar')
plt.title('Avg Spending by Customer Type')
plt.show()
```



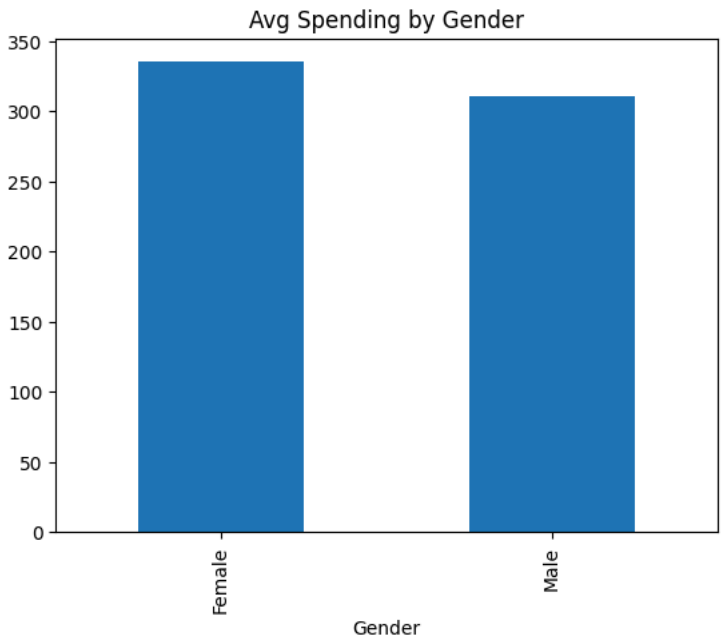
Insight: Customer Spending

Members spend more on average compared to normal customers.

Loyalty programs are effective in increasing spending.

Recommendation: Encourage more customers to become members.

In [ ]:
```
#which gender spending more
gender_spending = data.groupby('Gender')['Total'].mean()
gender_spending.plot(kind='bar')
plt.title('Avg Spending by Gender')
plt.show()
```

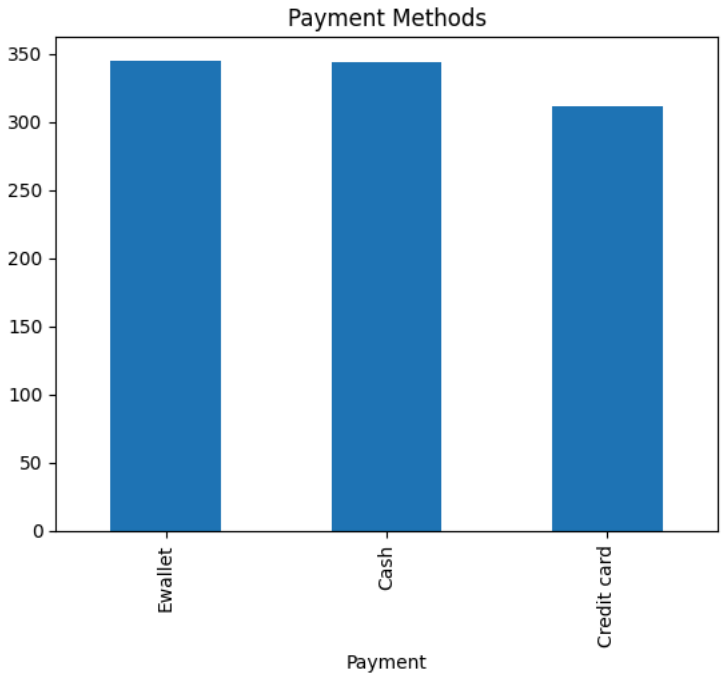


Insight: Gender Spending

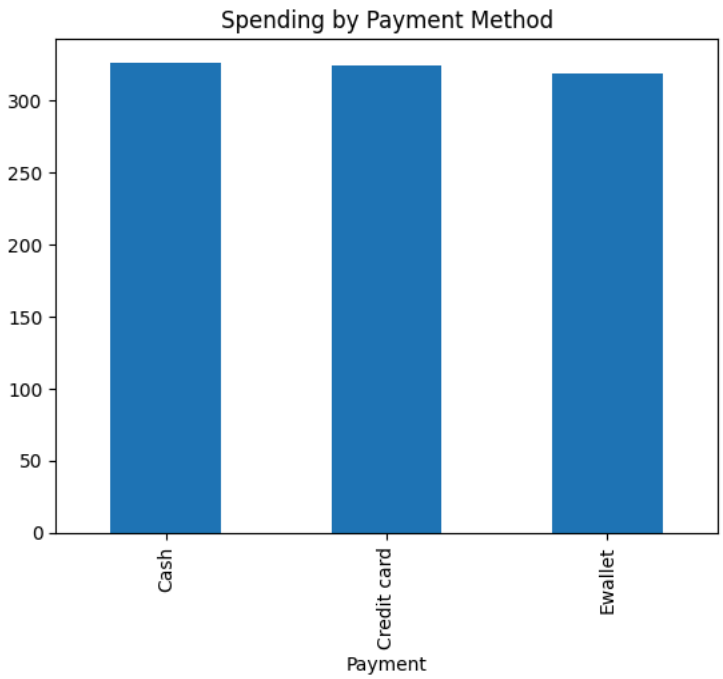
Female customers spend slightly more than male customers.

This suggests that marketing strategies could be tailored towards female customers.

In [ ]:
```
#most used payment method
payment_count = data['Payment'].value_counts()
payment_count.plot(kind='bar')
plt.title('Payment Methods')
plt.show()
```



In [ ]:
```
#spending
payment_spending = data.groupby('Payment')['Total'].mean()
payment_spending.plot(kind='bar')
plt.title('Spending by Payment Method')
plt.show()
```



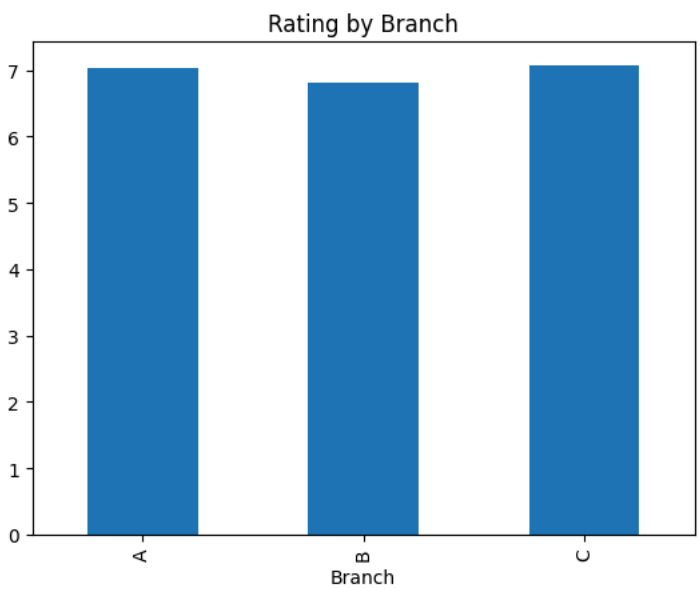
Insight: Payment Methods

E-wallet and cash are the most commonly used payment methods.

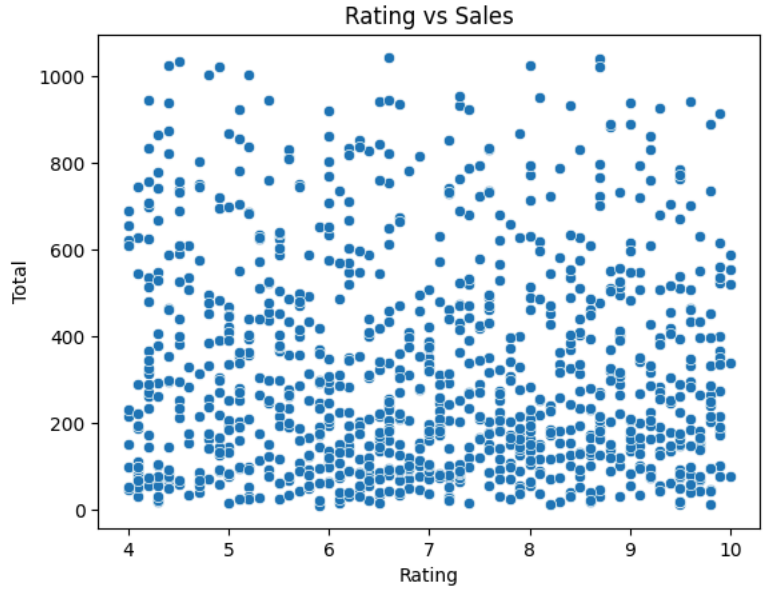
Digital payments are popular among customers.

Recommendation: Provide incentives for e-wallet usage to increase convenience.

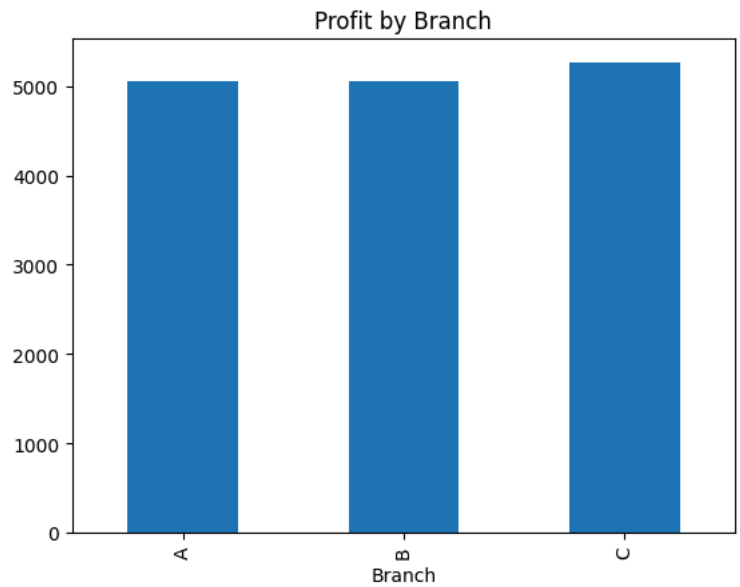
In [ ]:
```
#rating per branch
rating_branch = data.groupby('Branch')['Rating'].mean()
rating_branch.plot(kind='bar')
plt.title('Rating by Branch')
plt.show()
```



In [ ]:
```
#rating
sns.scatterplot(x='Rating', y='Total', data=data)
plt.title('Rating vs Sales')
plt.show()
```



In [ ]:
```
#profit by branch
profit_branch = data.groupby('Branch')['gross income'].sum()
profit_branch.plot(kind='bar')
plt.title('Profit by Branch')
plt.show()
```



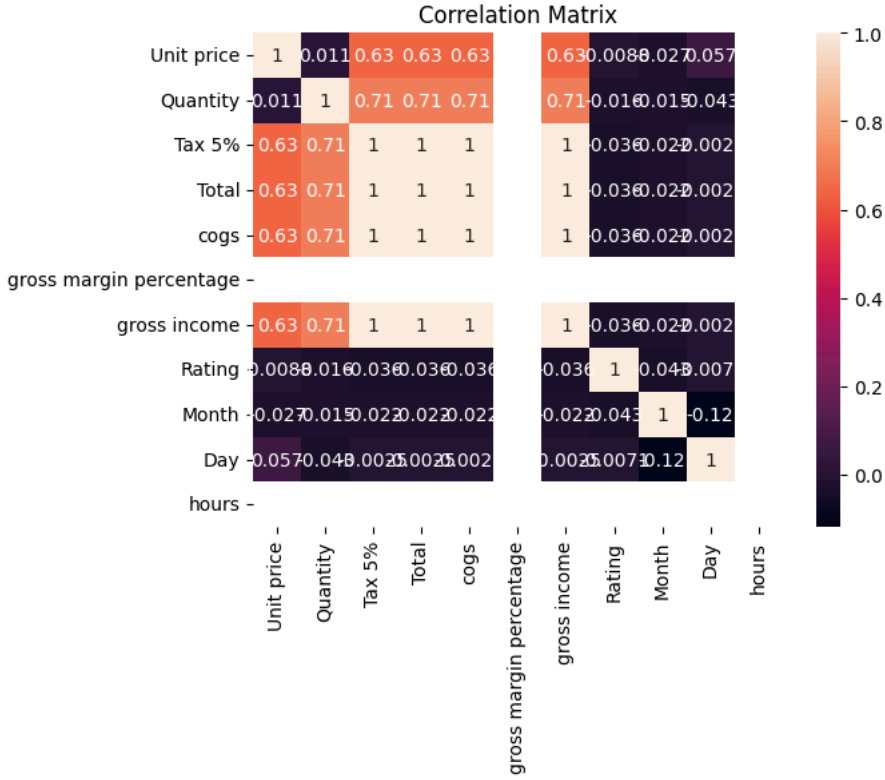
Insight: Profit by Branch

Branch C generates the highest profit.

This aligns with its high sales performance.

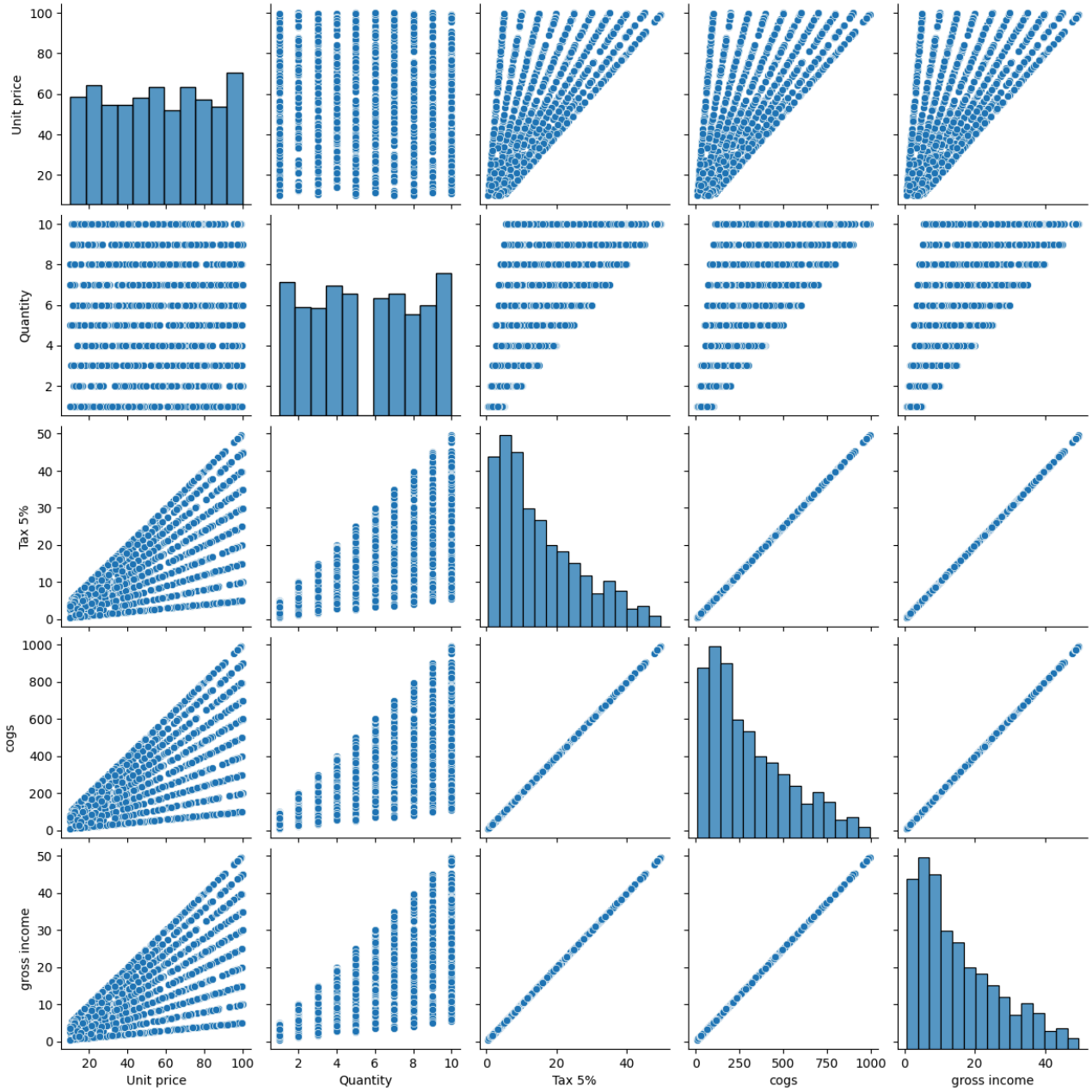
Recommendation: Study pricing and operations of Branch C.

In [ ]:
```
#correlation
sns.heatmap(data.corr(numeric_only=True), annot=True)
plt.title('Correlation Matrix')
plt.show()
```



In [ ]:

```
features = ['Unit price', 'Quantity', 'Tax 5%', 'cogs', 'gross income']
sns.pairplot(data[features])
plt.show()
```



Key Findings

- Naypyitaw is the top-performing city in terms of revenue
- Branch C leads in both sales and profit
- Electronic accessories have the highest sales volume
- Food and beverages generate the highest profit
- Members spend more than normal customers
- Female customers spend slightly more than males
- E-wallet and cash are the most used payment methods

Conclusion

The analysis shows that sales and profitability are influenced by multiple factors including location, product line, and customer type.

By applying targeted strategies, the supermarket can improve revenue, customer satisfaction, and overall business performance.

Recommendations

- Focus expansion strategies in high-performing cities like Naypyitaw
- Increase stock for high-performing product lines
- Promote membership programs to increase revenue
- Improve performance of low-performing branches
- Offer incentives for digital payments
- Optimize staffing during high-sales periods